

SQL ASSESSMENT

1. Table: `Users`

Tracks customer profile registration data and location profiles.

Structure:

```
CREATE TABLE Users (  
    UserID INT PRIMARY KEY,  
    UserName VARCHAR(50),  
    City VARCHAR(50),  
    AccountType VARCHAR(20) -- 'Regular' or 'Premium');
```

Sample Data:

```
INSERT INTO Users VALUES (1, 'Aman Verma', 'Delhi', 'Premium');  
INSERT INTO Users VALUES (2, 'Riya Sen', 'Mumbai', 'Regular');
```

2. Table: `Restaurants`

Tracks restaurant vendors partnered with the platform.

Structure:

```
CREATE TABLE Restaurants (  
    RestaurantID INT PRIMARY KEY,  
    RestaurantName VARCHAR(100),  
    Cuisine VARCHAR(50),  
    Rating DECIMAL(2,1));
```

Sample Data:

```
INSERT INTO Restaurants VALUES (101, 'Spice Symphony', 'North Indian',  
4.5);  
INSERT INTO Restaurants VALUES (102, 'Pizza Express', 'Italian', 3.9);
```

3. Table: `Orders`

Tracks transaction metrics generated by users at specific restaurants.

Structure:

```
CREATE TABLE Orders (  
    OrderID INT PRIMARY KEY,  
    UserID INT,  
    RestaurantID INT,  
    BillAmount DECIMAL(10,2),  
    OrderDate DATE,  
    FOREIGN KEY (UserID) REFERENCES Users(UserID),  
    FOREIGN KEY (RestaurantID) REFERENCES Restaurants(RestaurantID));
```

Sample Data:

```
INSERT INTO Orders VALUES (501, 1, 101, 1200.00, '2026-07-15');
INSERT INTO Orders VALUES (502, 2, 102, 450.00, '2026-07-16');
```

4. Table: `Deliveries`

Tracks delivery agent logs linked to completed platform orders.

Structure:

```
CREATE TABLE Deliveries (
    DeliveryID INT PRIMARY KEY,
    OrderID INT,
    DeliveryStatus VARCHAR(20), -- 'Delivered', 'Cancelled', 'In-Transit'
    DeliveryTimeMinutes INT,
    FOREIGN KEY (OrderID) REFERENCES Orders(OrderID));
```

Sample Data:

```
INSERT INTO Deliveries VALUES (901, 501, 'Delivered', 25);
INSERT INTO Deliveries VALUES (902, 502, 'Delivered', 42);
```

SECTION A:**Question 1**

- **Problem Statement:** Write a query to display the `UserName` of the customer, the `RestaurantName` they ordered from, and the `BillAmount` for all orders.
- **Tables to Use:** `Users`, `Restaurants`, `Orders`

Question 2

- **Problem Statement:** Fetch a unique list of all restaurant names that have successfully had orders cooked and assigned to a delivery tracking record (even if the delivery was later cancelled).
- **Tables to Use:** `Restaurants`, `Orders`

Question 3

- **Problem Statement:** Display all `OrderID` logs along with their corresponding user names where the order took more than 35 minutes to deliver.
- **Tables to Use:** `Users`, `Orders`, `Deliveries`

Question 4

- **Problem Statement:** Find the total money spent (`BillAmount`) on food delivery orders by each individual `UserName`. Display the user's name and their total spend amount.
- **Tables to Use:** `Users`, `Orders`

SECTION B:

Question 5

- **Problem Statement:** The business intelligence team needs to see a comprehensive dispatch chart. Write a query to list **every user name** registered on the system along with the total number of orders they have placed. If a user has placed 0 orders, their name must still appear in the list with an order count of 0.
- **Tables to Use:** `Users`, `Orders`

Question 6

- **Problem Statement:** Write a query to find the names of restaurants that have generated a total cumulative revenue (sum of `BillAmount`) greater than ₹5,000 from users who live specifically in the city of 'Delhi'.
- **Tables to Use:** `Users`, `Restaurants`, `Orders`

Question 7

- **Problem Statement:** Identify structural system inefficiencies. Write a query to find the names of all restaurants that have suffered from 'Cancelled' orders, along with the total count of cancelled orders each restaurant faced.
- **Tables to Use:** `Restaurants`, `Orders`, `Deliveries`

SECTION C:

Question 8

- **Problem Statement:** Write a query to find the `UserName` and `Email/City` profiles of users who have spent more money on a *single order* than the average order value computed across the entire platform.
- **Tables to Use:** `Users`, `Orders`

Question 9

- **Problem Statement:** Generate a platform-wide operational efficiency audit report. Write a query to calculate the average delivery time (`DeliveryTimeMinutes`) for each distinct `Cuisine` category offered on the platform, but only include cuisines where the restaurant's average rating is strictly above 4.0.
- **Tables to Use:** `Restaurants`, `Orders`, `Deliveries`

Question 10

- **Problem Statement:** Write a query to rank restaurants within each unique `Cuisine` category based on the total number of orders they have received (highest orders ranked 1st). The output should show the Cuisine, Restaurant Name, Order Count, and their Rank.
- **Tables to Use:** `Restaurants`, `Orders`